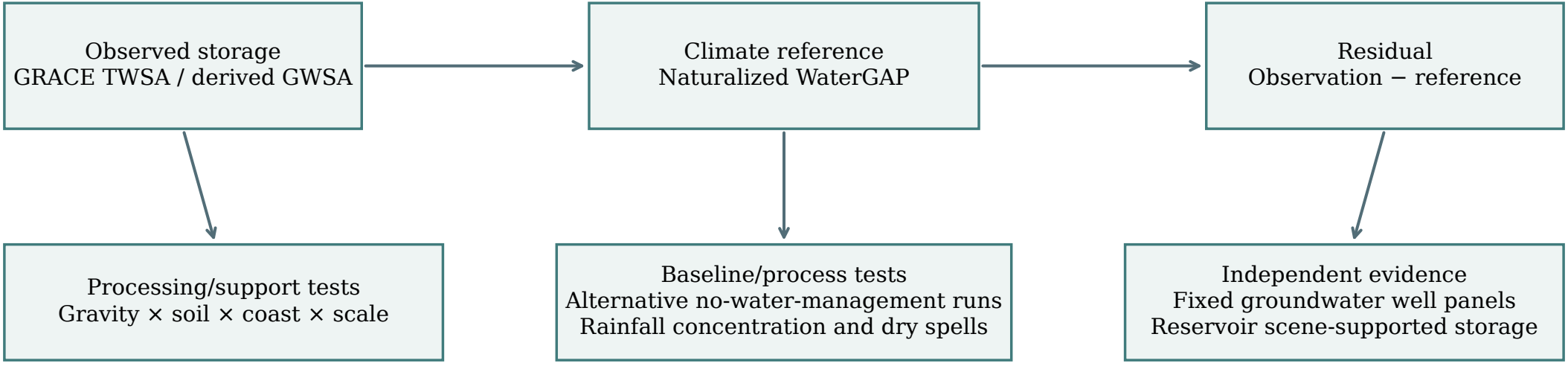
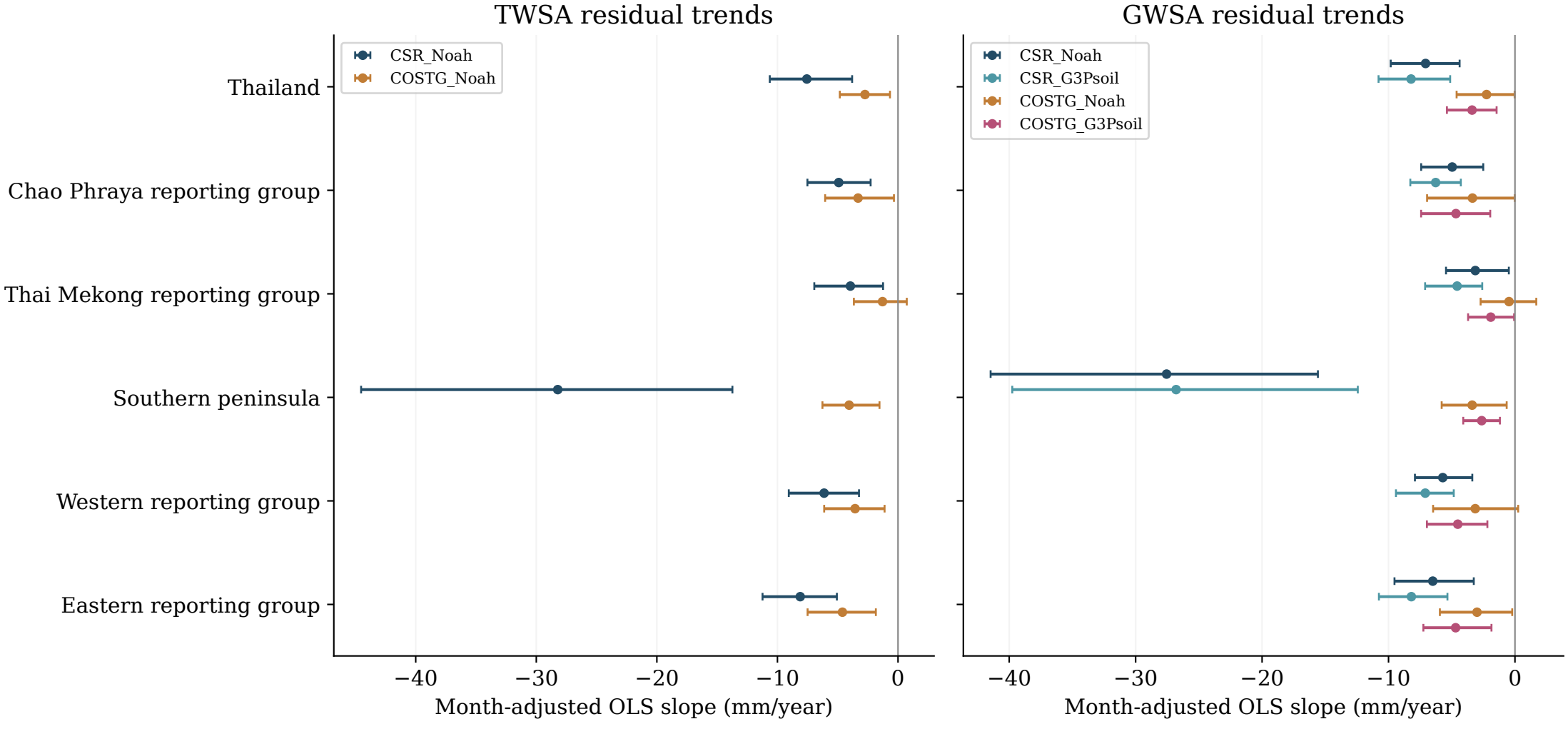
A direct-model framework for testing attribution limits



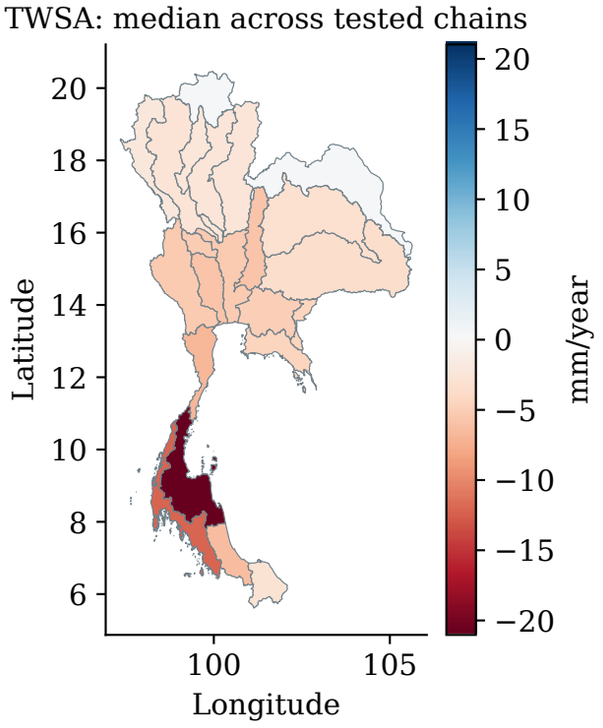
The analysis tests robustness and physical consistency; it does not identify all residual storage as human-caused.
Climate reconstruction and residence-time fitting are excluded from this design.

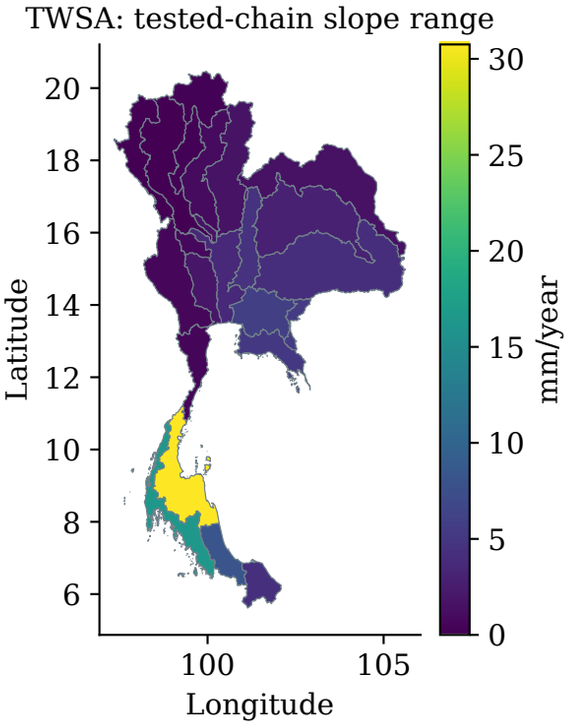
Do declines survive gravity and soil-component choices?

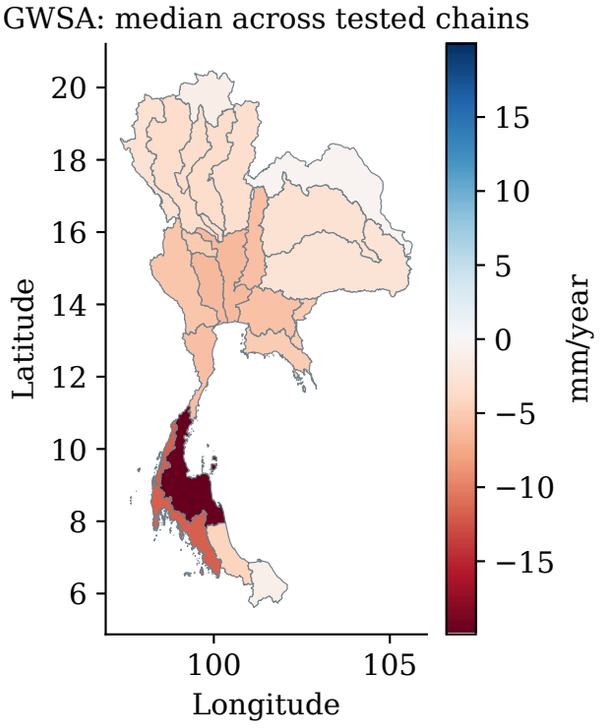


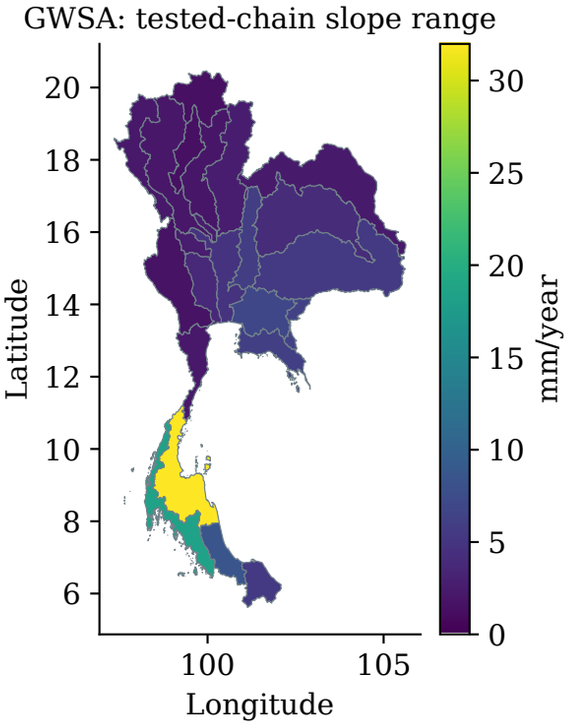
Intervals: conditional two-year calendar-block wild regression; they do not include all satellite/model uncertainty.
Regional groups aggregate reporting basins, not new independent GRACE measurements; TWSA does not depend on soil choice.

Residual trend magnitude and its processing sensitivity



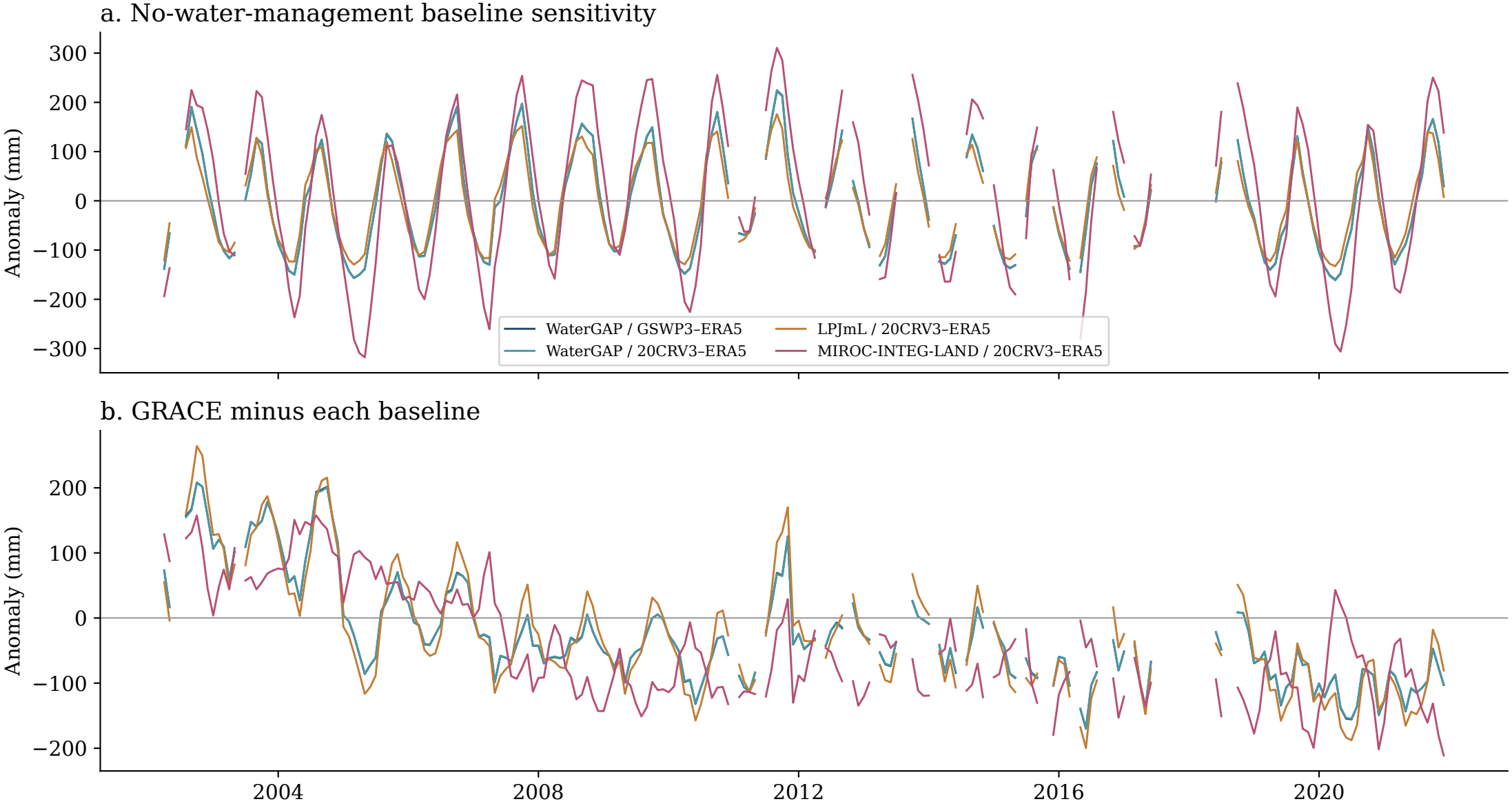






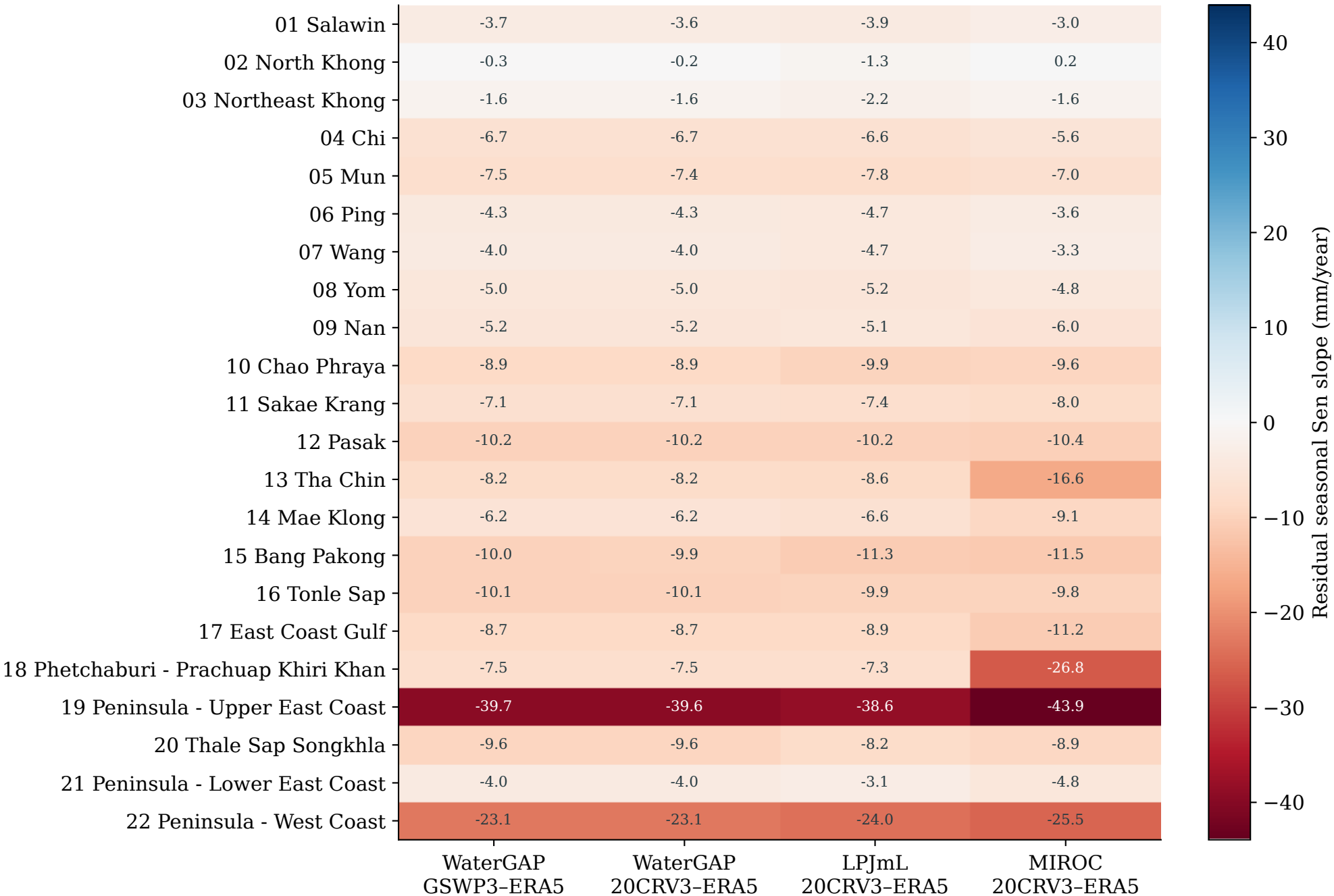
Median/range across two gravity products and, for GWSA, two soil estimates. Other subtraction terms are held fixed.
A narrow range within this tested set is not proof of small total uncertainty; basin-scale GRACE support remains coarse.

How strongly does the storage difference depend on the reference?



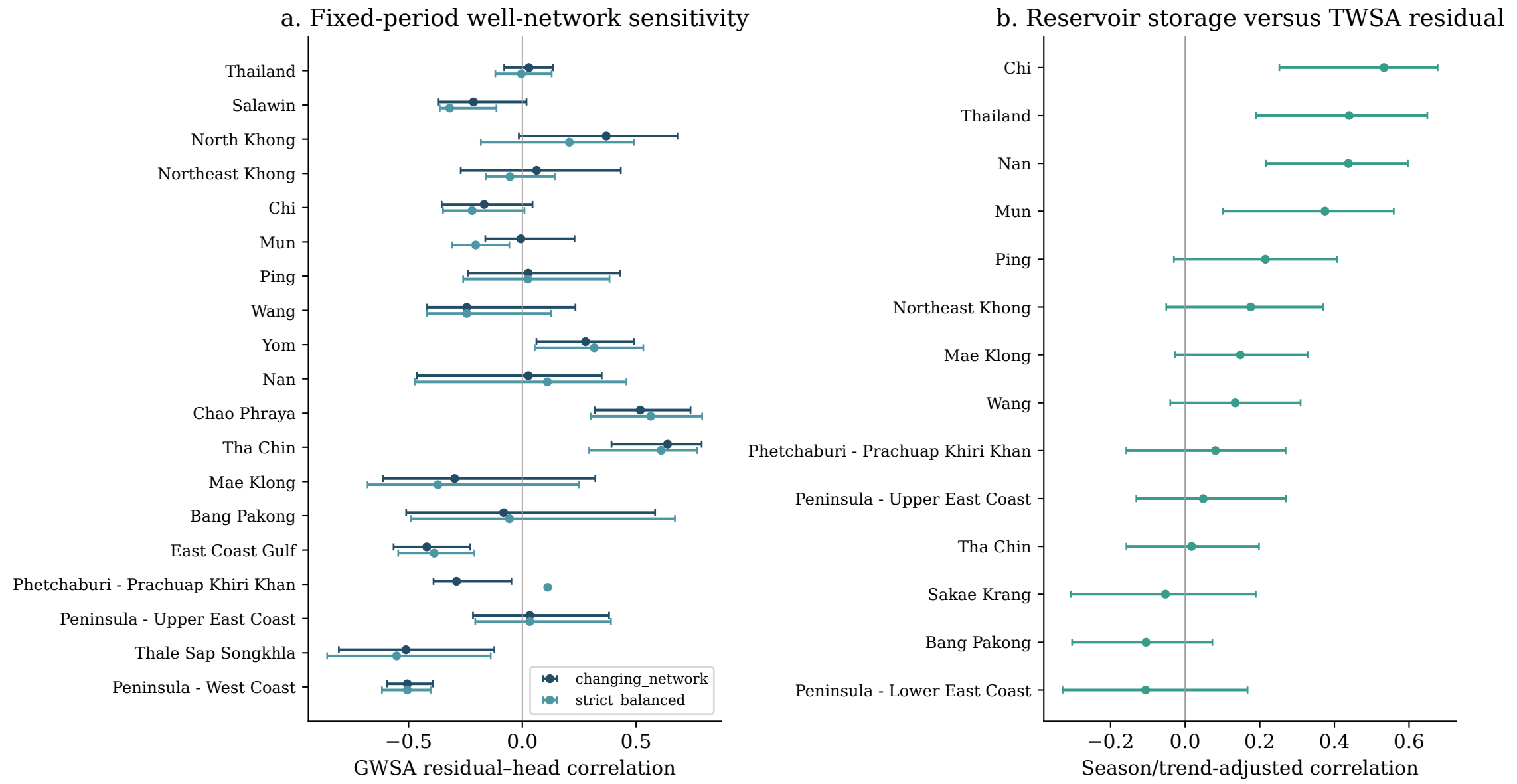
Exact common 2002–2021 support and 2004–2009 reference. LPJmL/MIROC are structural diagnostics, excluded from full-TWS evidence flags.
Their groundwater accounting is not verified as equivalent to WaterGAP; nowatermgt can retain other historical human forcings.

Basin storage differences across four reference simulations



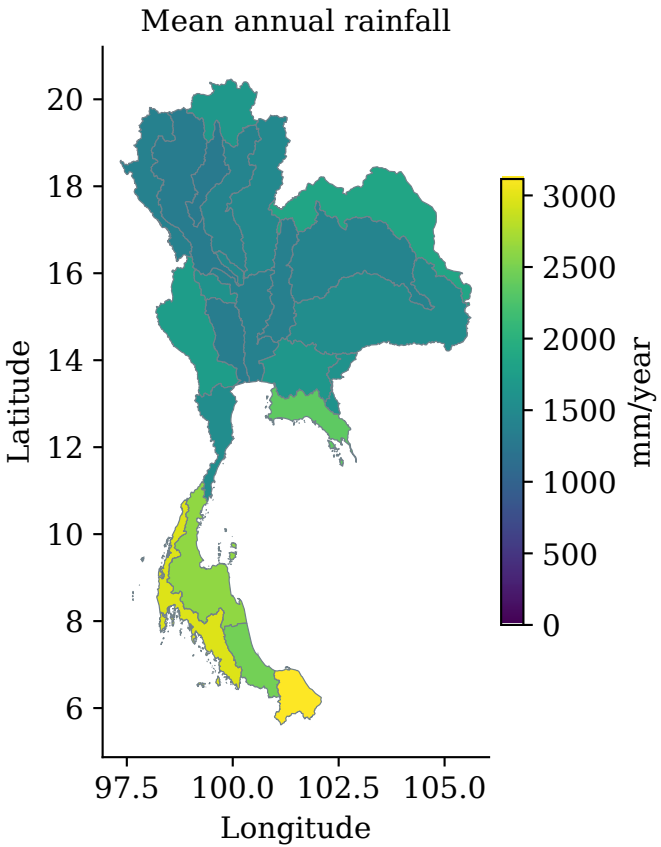
TWSA differences, common 2002–2021 support. LPJmL/MIROC are structural diagnostics, not verified equivalent full-TWS baselines.
Only the two WaterGAP variants enter baseline-robustness flags. Differences combine forcing, model structure and experiment definitions.

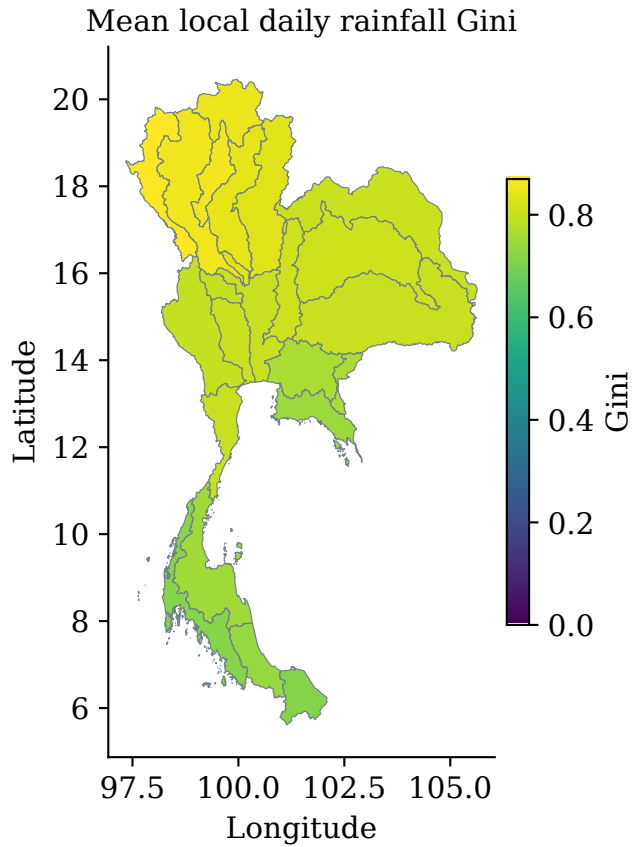
What do independent observations support?

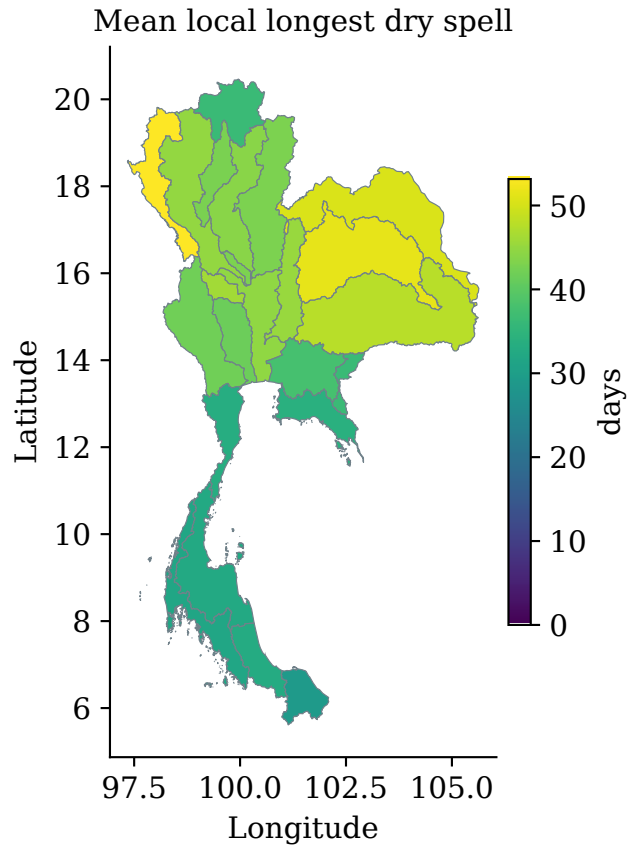


Wells: 2013–2022, ≥8 paired years; station-bootstrap intervals conditional on observed years (changing/strict panels).
Reservoirs: ≥24 scene-supported months; year-cluster intervals. Correlations are not causal attribution or storage-volume validation.

Rainfall amount, concentration and dry spells: 2003–2023

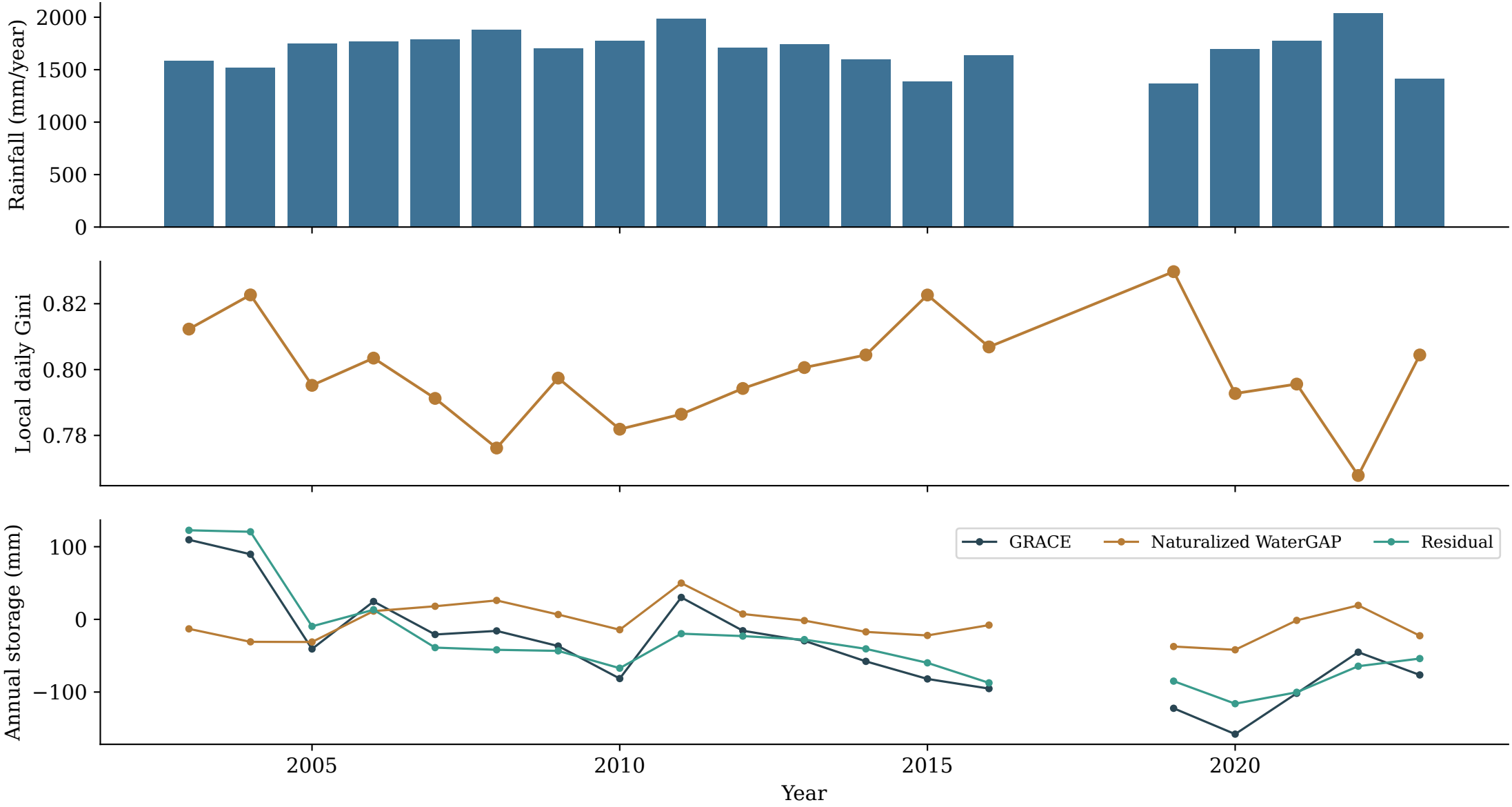






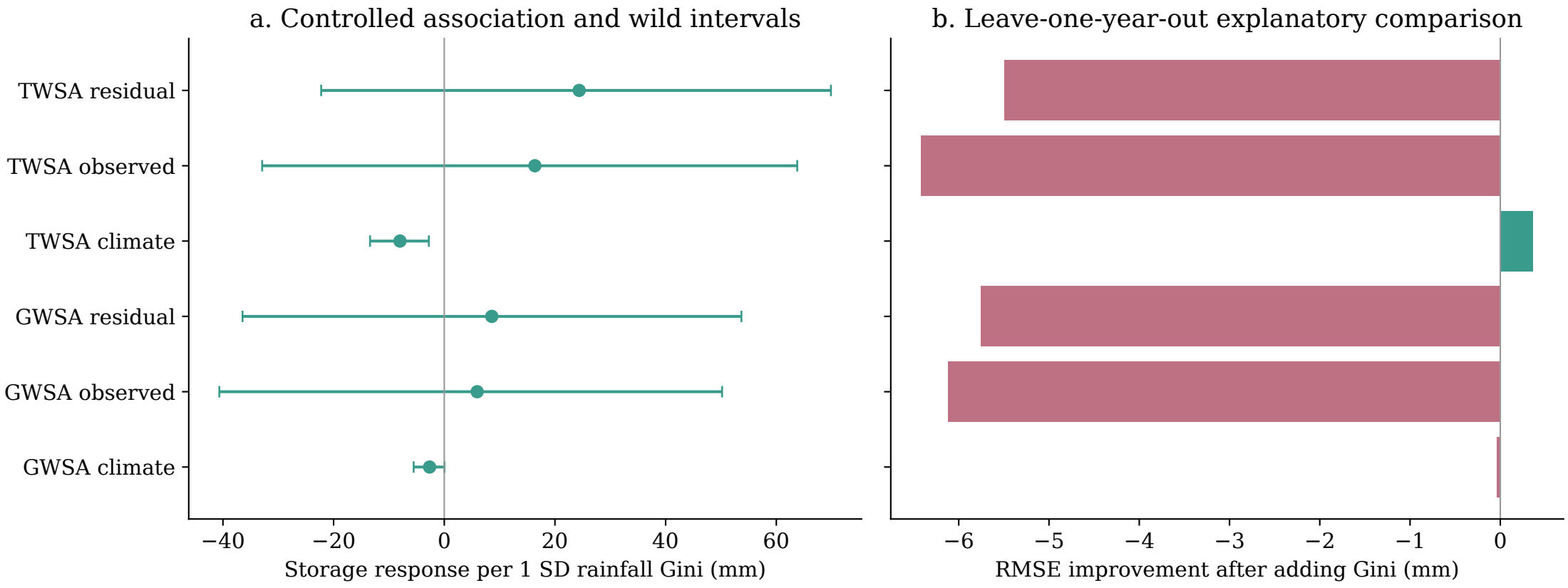
CHIRPS v2 daily at native 0.05° pixel centres, cosine-latitude area weights. Higher Gini means more concentrated daily rainfall.
Local indices are calculated per grid cell before basin averaging; dry days have rainfall <1 mm. Daily CHIRPS is a derived rainfall product.

National rainfall concentration and storage variability



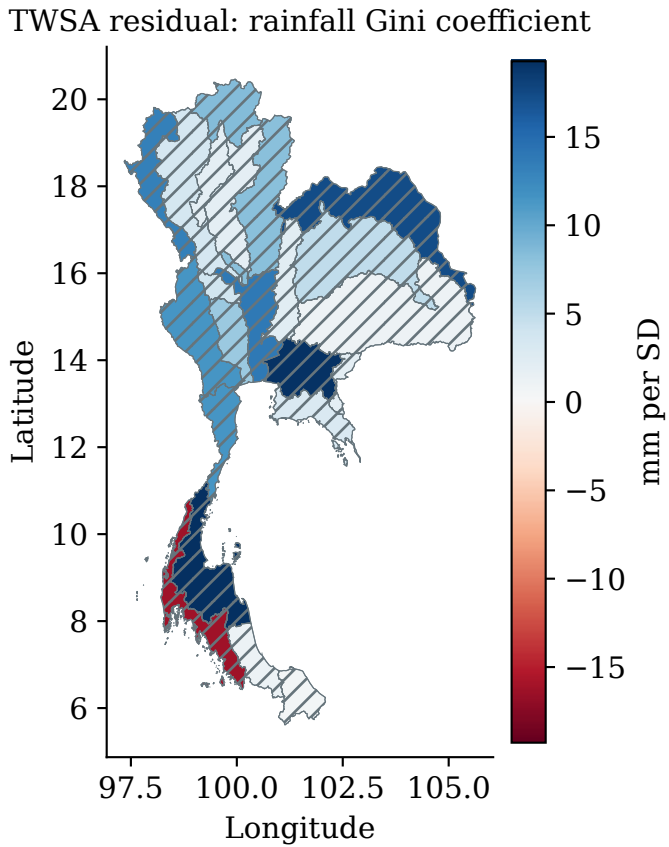
Storage: mean deseasonalized anomaly on common dates, ≥8 observed months per year; no mission-gap filling.
Rainfall and Gini use full calendar years. This display is descriptive; the controlled analysis also includes annual PET and time.

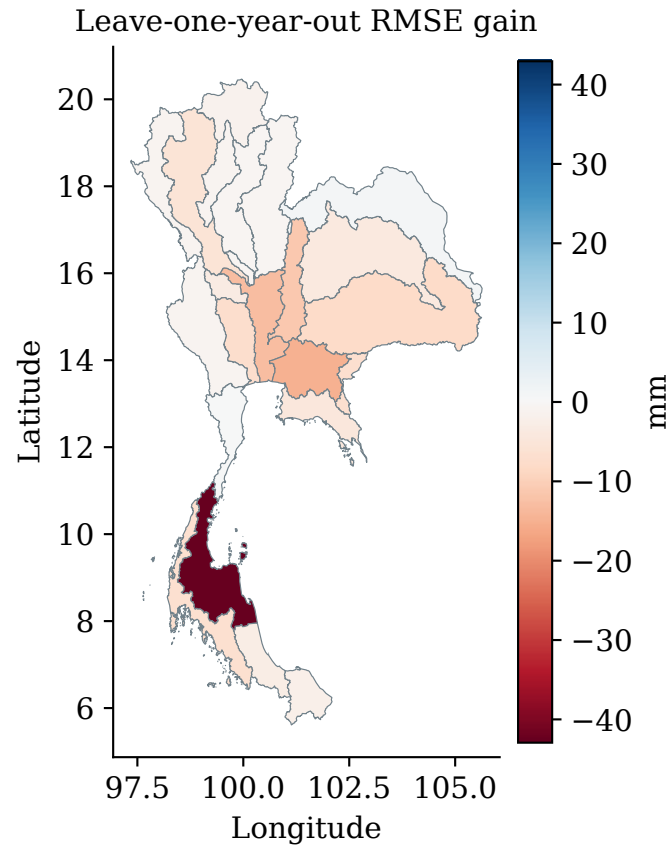
Does rainfall concentration add information beyond rainfall totals?



Controls: annual precipitation, annual potential ET and linear time; main predictor: area-mean local daily Gini.
Two-year wild intervals; small annual samples. Positive RMSE gain means improvement. These are exploratory associations, not causal effects.

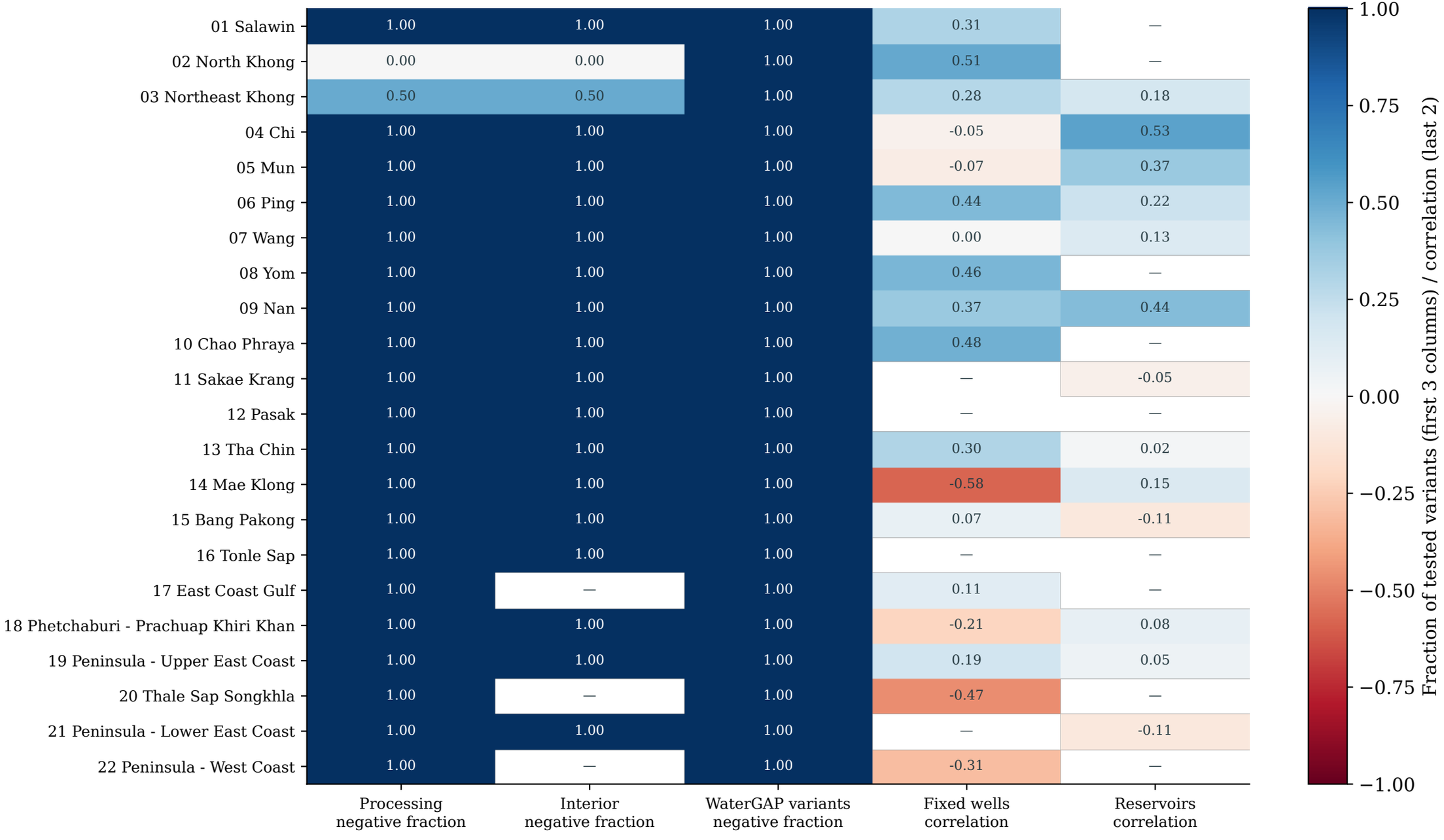
Spatial evidence for rainfall concentration in the residual





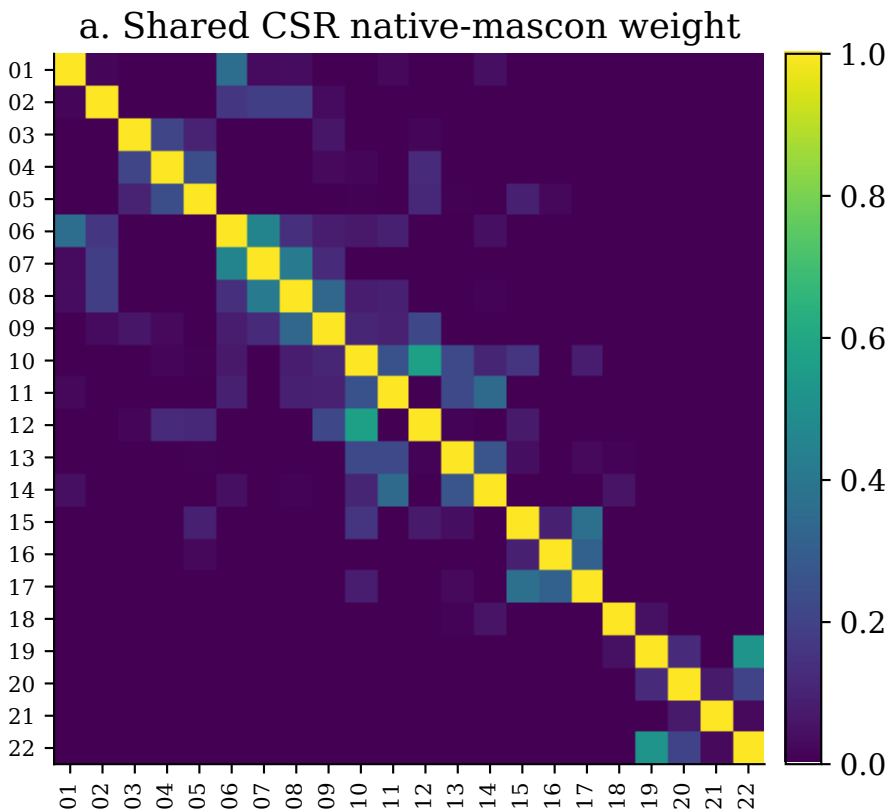
Hatching: q ≥ 0.05 after adjustment across all fixed basin tests (both storage variables, three responses, three rain metrics).
Basins share GRACE/climate signals. Improved explanatory fit alone does not establish a mechanism or anthropogenic influence.

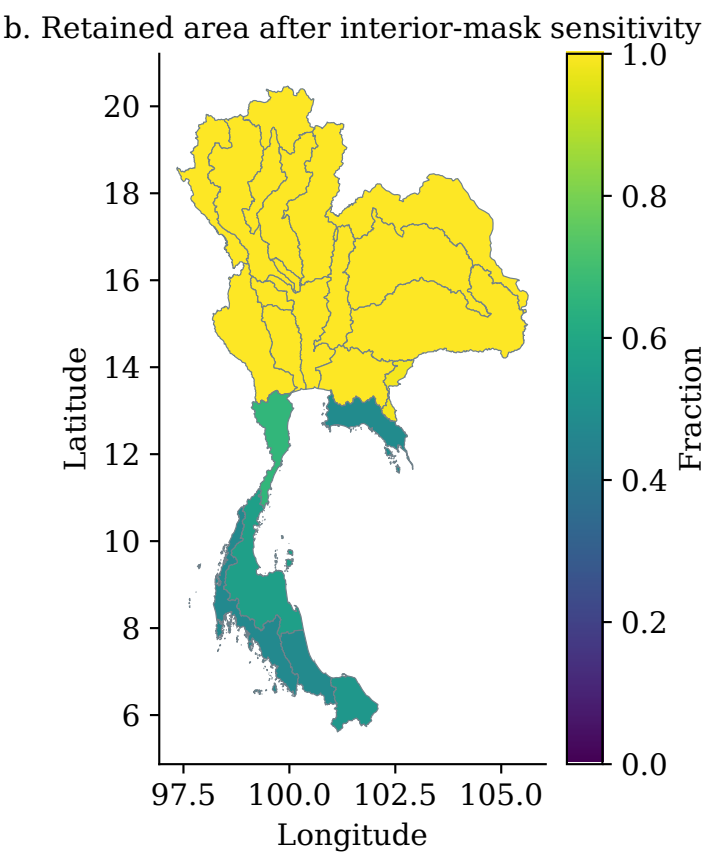
Basin evidence: robust declines, sparse support and disagreement



Fractions describe tested variants, not probabilities; baseline flags use only two WaterGAP variants, not the structural diagnostics.
Correlations describe consistency, not causality. Missing evidence remains blank; no overall confidence score is assigned.

Why 22 reporting basins are not 22 independent satellite regions





Shared native weights describe overlapping sampling, not a full GRACE error covariance or independent effective sample size.
Interior sensitivity removes a 0.25° grid-cell border next to provider ocean masks; it is a stress test, not a validated coastal correction.